

Code-Layout, Readability and Reusability

Date: October 3, 2026

Team ID: SWTID-2026-7578

Project Name: Implement Client Script & UI Policy (Incident)

Maximum Marks: 5 Marks

**Code-Layout, Readability and Reusability:**  
This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

CODE LAYOUT CHECKLIST

| S.No | Code Quality Parameter    | Description                                   | Followed | Remarks                                                                                                                            |
|------|---------------------------|-----------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Consistent Indentation    | Uniform spacing/tabs used throughout the code | Yes      | Standard 2-space JavaScript indentation applied across scripts.                                                                    |
| 2    | Proper File Structure     | Files and folders are logically organized     | Yes      | Modularized into discrete Client Scripts and UI Policy Actions in Update Sets.                                                     |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly       | Yes      | Clear variable names like <code>impactVal</code> , <code>assignedTo</code> , and <code>g_form</code> handles used.                 |
| 4    | Function/Method Names     | Functions are descriptively named             | Yes      | Standard ServiceNow entry functions ( <code>onChange</code> , <code>onSubmit</code> , <code>onCellEdit</code> ) strictly followed. |
| 5    | Code Comments             | Inline and block comments explain logic       | Yes      | JSDoc block comments explain parameters, trigger fields, and error handling logic.                                                 |
| 6    | Modular Design            | Code is split into reusable functions/modules | Yes      | Validation logic isolated into dedicated, reusable helper functions.                                                               |
| 7    | No Redundant Code         | Duplicate or unused code is removed           | Yes      | Dead code eliminated; shared validation logic consolidated into single calls.                                                      |
| 8    | Error Handling            | Exceptions and errors are handled gracefully  | Yes      | Uses <code>g_form.showErrorBox()</code> and returns <code>false</code> on invalid form submissions.                                |

REUSABLE COMPONENTS / MODULES

| S.No | Component / Module Name      | Language / Technology              | Where Reused                                            | Reusability Level |
|------|------------------------------|------------------------------------|---------------------------------------------------------|-------------------|
| 1    | Mandatory Field Validator    | JavaScript (ServiceNow Client API) | <code>onSubmit</code> Incident Form & Task Forms        | High              |
| 2    | Impact-Urgency Auto-Sync     | JavaScript (GlideForm API)         | <code>onChange</code> Impact Listener across ITSM forms | High              |
| 3    | Grid List Cell Editor Lock   | JavaScript (onCellEdit API)        | Incident, Problem & Change List Views                   | High              |
| 4    | High-Impact UI Policy Action | ServiceNow UI Policy Engine        | Incident Form Layout Rules                              | High              |
| 5    | Custom Field Alert Banner    | JavaScript / Client API            | All ITSM Task Forms                                     | High              |

OVERALL CODE QUALITY ASSESSMENT

| Aspect                   | Rating (1-5) | Comments                                                                        |
|--------------------------|--------------|---------------------------------------------------------------------------------|
| Code Layout & Structure  | 5 / 5        | Clean structure adherence to ServiceNow client scripting standards.             |
| Readability              | 5 / 5        | Exemplary variable naming and descriptive callback parameters.                  |
| Reusability              | 5 / 5        | Scripts and UI policy rules are decoupled and ready for export via Update Sets. |
| Documentation / Comments | 5 / 5        | Comprehensive block and inline JSDoc comments detailing conditions.             |
| Overall Score            | 5 / 5        | Excellent codebase quality meeting enterprise standards.                        |